

page1/2
using s3a

voidsolid 20151122 09AM

```
OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMZFPPKWGDZXTJCDIGKUHUAUEKCAR pt
-----ELYOIECBAQK-----
KRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLM s1
KZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCB s2
IMUXRTADGJNVZYOBELQWKPSCFIMUXRTADGJNVZYOBELQWKPSCFIMUXRTADGJNVZYOBELQWKPSCFIMUXRTADGJNVZYOBEL s3a
```

```
OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMZFPPKWGDZXTJCDIGKUHUAUEKCAR
-----BERLINLOCK----- pt
OAXXLKMX00ESVNLYZUJIPLUTXUQYZJCUKZWUGNBNOJINAOTRSLKJNAEYLBEPXBJJBTHRANCORJFRZWYG00CSMFTDUBWENJC
OCYTZDTFUWOYQNN SOPKRLAGXQNQTPVLOFGGAI SNTCVWPNNQNTYIKMWGNFPVRUMTHJDNQCLOWUNDBZZZSNGIUQBTXSNBZIZXN
JJOPQRLUKXYUCTKCTXLHKFHUGBOESUFVZVMHTBUTEFUMQSYZQRUTEZKFHPJOSVIWPRLVNTBMSKQARHDSMBSBPGOUZBNSMXND
```

using s3b

```
OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMZFPPKWGDZXTJCDIGKUHUAUEKCAR
-----ELYOIECBAQK-----
KRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLM s1
KZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCBASOTPYRKZXWVUQNMLJIHGFEDCB s2
IMUXRTADGJNVZRTADGJNVZYOBELQWKPSCFIMUXKPSCFIMUXRTADGJNVZYOBELQWKPSCFIMUXRTADGJNVZRTADGJ s3b
```

```
OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMZFPPKWGDZXTJCDIGKUHUAUEKCAR
-----BERLINLOCK----- pt
OAXXLKMX00ESVNLYZUJIPLUTXUQYZJCUKZWUGNBNOJINAOTRSLKJNAEYLBEPXBJJBTHRANCORJFRZWYG00CSMFTDUBWENJC
OCYTZDTFUWOYQNN SOPKRLAGXQNQTPVLOFGGAI SNTCVWPNNQNTYIKMWGNFPVRUMTHJDNQCLOWUNDBZZZSNGIUQBTXSNBZIZXN
JJOPQRLUKXYUCORDOZMIRGIVHCSFAVGWKWNIOCVSGHWQVBTRVPWSGRYHJOMABXLZORLVNTBMSKQARHDSMBSBPGOUZBQANZQE
```

it is trivial how to derive s1 and s2. All alingments are also trivial.

page1/2

page2/2

voidsolid 20151122 09AM

how s3a and s3b are derived:

KRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZKRYPTOSABCDEFGHIJLMNQUVWXZ

KRY
PTO
SAB
CDE
FGH
IJL
MNQ
UVW
XZK
RYP
TOS
ABC
DEF
GHI
JLM
NQU
VWX
ZKR
YPT
OSA
BCD
EFG
HIJ
LMN
QUV
WXZ

s3a: if rotate left first row only

YOB EHLQWKPSCFIMUXRTADGJNVZ

s3b: if rotate left and all 3 rows

YOB EHLQWKPSCFIMUXRTADGJNVZ RTADGJNVZ YOB EHLQWKPSCFIMUXK PSCFIMUXRTADGJNVZ YOB EHLQW

page2/2